# SMART INDIA HACKATHON 2026

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## TITLE PAGE

- Problem Statement ID – SIH26171
- Problem Statement Title- On-device Visual Perception for Light-weight Browser Agents
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal) - StrawHats

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# IDEA TITLE

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## ✻ Proposed Solution

**On-Device Privacy Gateway:** A lightweight browser layer that understands the interface locally, ensuring raw screens, credentials and personal information never leave the device unnecessarily.

**Adaptive Visual Perception:** Combines DOM, accessibility signals and local Vision/OCR, automatically falling back to visual understanding for Canvas, WebGL, PDFs and non-structured interfaces.

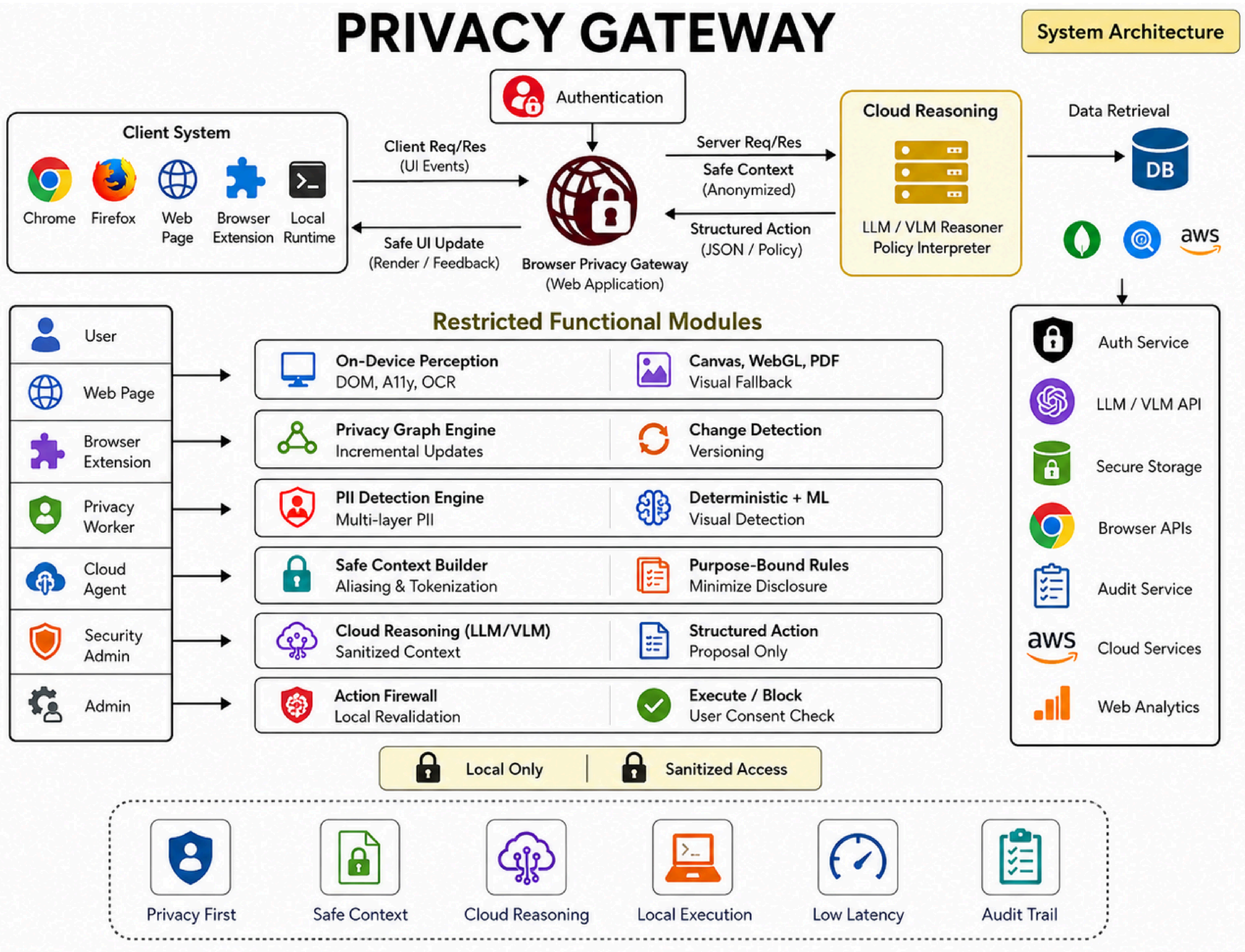
**Incremental Privacy Graph:** Maintains a continuously updated privacy-aware representation of the page and processes only what changes, cutting repeated computation, bandwidth, latency and resource usage.

**Multi-Layer PII Intelligence:** Detects passwords, emails, phones, cards, OTPs, Aadhaar, PAN, UPI, addresses and contextual PII using browser semantics, deterministic validators, compact on-device ML and visual detection.

**Purpose-Bound Privacy:** Sensitive data is replaced with safe aliases, and only the information actually needed for the task is kept or exposed.

**Cloud Reasons, Browser Decides:** The LLM only sees sanitized context and suggests an action. The browser checks the site, target, page state, sensitivity and user permission before doing anything.

**Fast by Design:** We update only what changes, reuse cached results and avoid constant vision scans, keeping the normal privacy path under 50 ms p95 while uncertain cases stay blocked instead of leaking.



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# TECHNICAL APPROACH

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## ❖ Technology Stack

**Frontend :** React Js (Vite), TypeScript, Tailwind CSS, Framer-Motion, Zustand, React Query, Chart.js, React Hook Form.

**Browser Extension :** TypeScript, Manifest V3, Content Scripts, Background Service Worker, Offscreen Documents, Web Workers.

**On-Device Engines :** DOM & Accessibility Parser, OCR (Tesseract.js), Vision (ONNX Runtime Web), PII Detection (ML + Heuristics), Privacy Graph Engine, Change Detection.

**AI / LLM Layer :** Llama 3.1 8B / Gemma 2 9B (via OpenRouter), Vision Model (On-device fallback), Embedding Model (Nomic Embed).

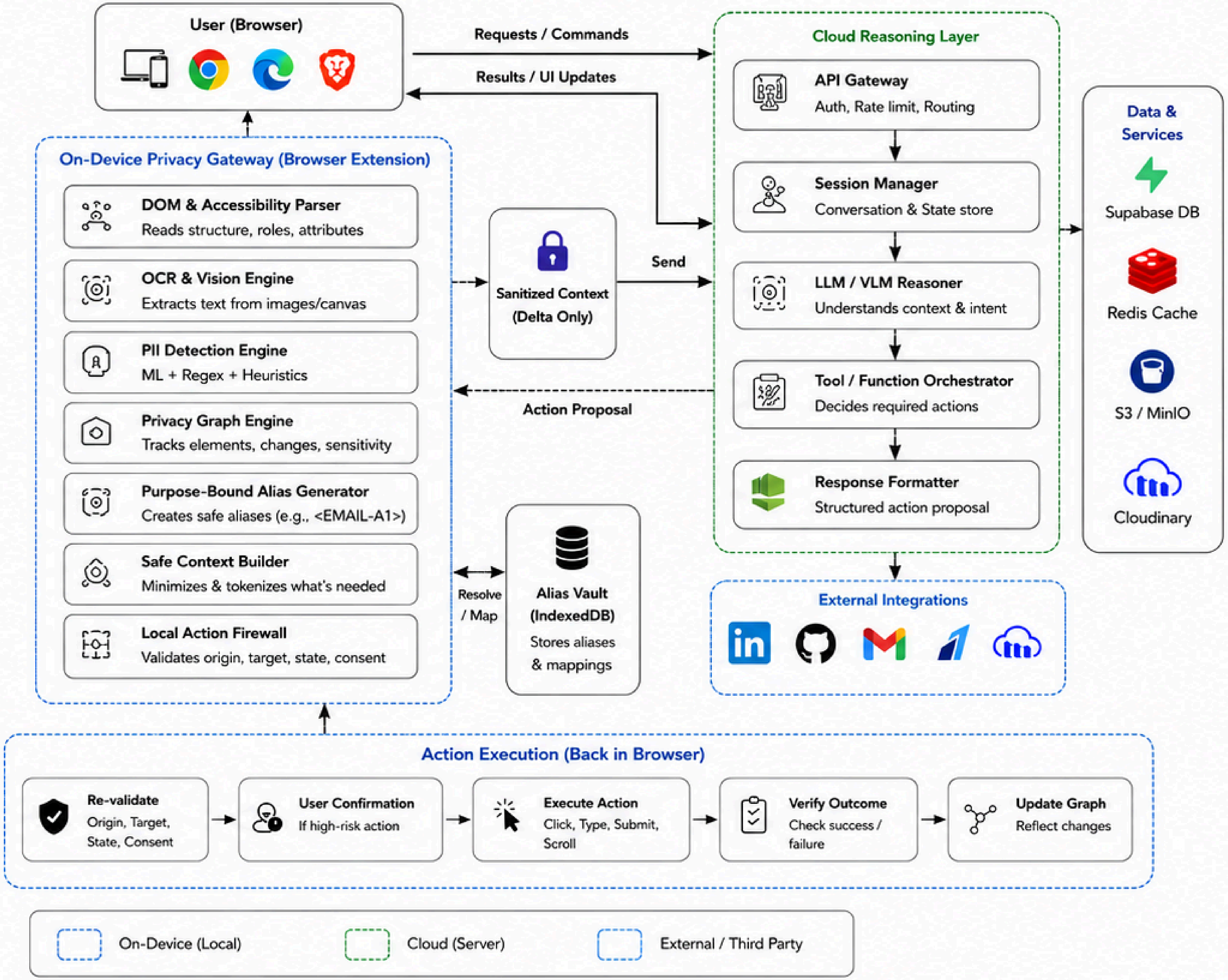
**Backend Services :** Node Js, Express Js, REST API, WebSocket, PostgreSQL, Redis, MinIO (S3), Supabase (Auth + DB).

**Security & Privacy :** AES-256-GCM Encryption, Alias Vault (IndexedDB), Origin Isolation, CSP, Permission Checks, RBAC.

**APIs & Integrations :** Browser APIs (DOM, A11y, Canvas, WebGL), LinkedIn API, GitHub API, Gmail API, Razorpay API, Cloudinary API.

**Deployment :** Vercel (Frontend), Render (API), Supabase Cloud, Cloudinary, Upstash Redis.

**Dev & Ops Tools :** Git, GitHub, Docker, Postman, Swagger, Sentry, Grafana, Prometheus, GitHub Actions (CI/CD).



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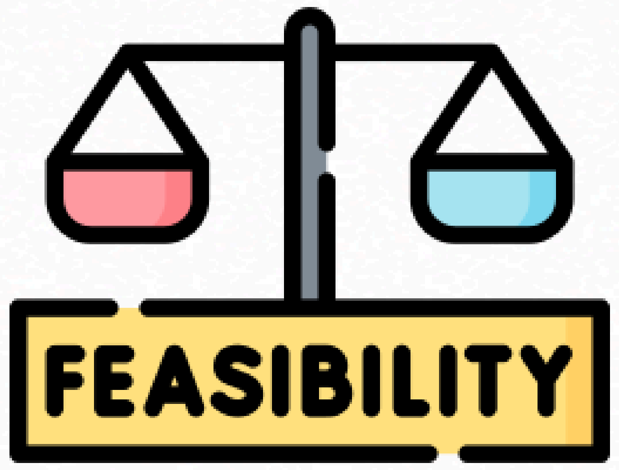
# FEASIBILITY AND VIABILITY

## Feasibility:

- Built using **proven browser capabilities** (DOM, Accessibility Tree, WebGPU, Service Workers, IndexedDB).
- **On-device ML inference** with ONNX Runtime Web and WebAssembly for privacy sensitive processing.
- **Incremental processing** + caching ensures sub-50ms warm-path and efficient resource usage.
- **Privacy-by-design architecture**: data stays on-device, only sanitized context is shared.
- All components are **open-source technologies** and readily integrable within hackathon timeline.

## Challenges:

- **PII detection accuracy** across diverse UI/visual layouts and languages.
- Balancing **context minimization** with maintaining high task accuracy.
- **Resource constraints** on low-end devices (CPU, Memory, GPU availability).
- **Security risks**: prompt injection, malicious pages, and data exfiltration attempts.
- Ensuring **reliability** across dynamic web pages and rapid UI changes.




## Strategies:

- **Hybrid PII detection** (rules + ML + OCR) with continuous model improvement.
- **Purpose-bound minimization** using Privacy Graph and alias tokens for safe context sharing.
- **Adaptive processing pipeline** that scales with device capability and network conditions.
- Robust **Action Firewall** with origin validation, user confirmation, and audit logging.
- **Continuous evaluation** and benchmarking to optimize accuracy, latency, and resource usage.

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# IMPACT AND BENEFITS

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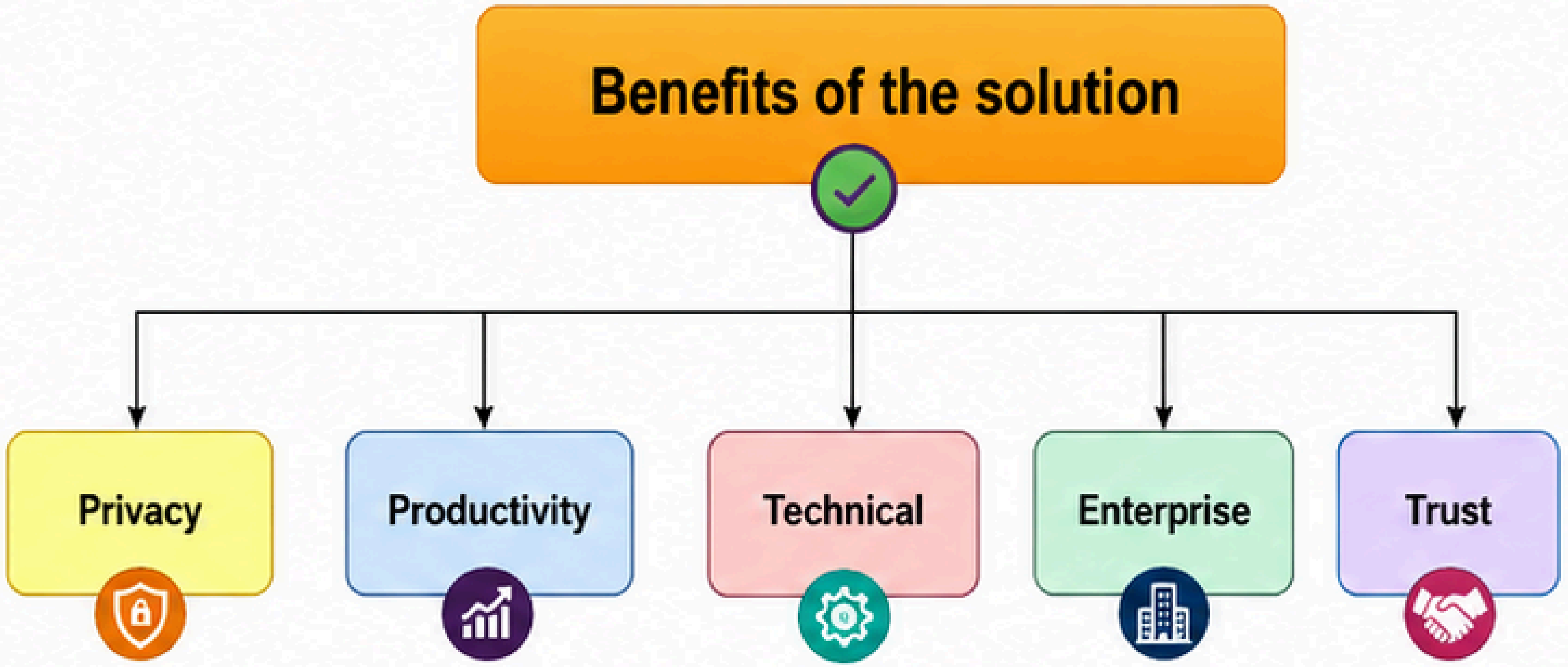


## Potential impact on the target audience:

- **End users:** Enables safe AI assistance for sensitive workflows such as login, checkout, form filling, support portals, and e-governance tasks without exposing raw screen data to the cloud.
- **Developers & agent builders:** Acts as a reusable privacy gateway, reducing the need to build custom redaction, consent, and safe-execution logic for every browser agent from scratch.
- **Enterprises & regulated sectors:** Unlocks adoption in finance, healthcare, education, HR, and government workflows where privacy, auditability, and controlled disclosure are major blockers.
- **AI ecosystem:** Encourages safer browser automation by introducing safe-context sharing, action validation, and local audit receipts as a practical trust layer for future agents.

## Benefits of the solution

**Privacy & Security:**

- Sensitive values stay on-device; only sanitized, anonymized context is shared with the cloud model.
- Task-scoped aliases let the agent use emails, cards, or addresses without ever seeing the real values.
- Local action firewall blocks prompt-injection, wrong-origin actions, and unsafe disclosures before execution.

**Technical:**

- Structure-first perception uses DOM and accessibility signals before invoking OCR or visual fallback, improving efficiency.
- Incremental Privacy Graph and delta updates reduce repeated processing, bandwidth usage, and warm-path latency.
- Adaptive fallback keeps coverage for Canvas, WebGL, PDFs, and visual-only interfaces when structure is unavailable.

**User & Workflow Benefits:**

- Users get powerful browser assistance without giving up control of private credentials, messages, or personal data.
- Risky actions can require explicit confirmation, while routine low-risk actions remain fast and seamless.
- Audit receipts make it clear what data stayed local, what was shared, and what action was executed.

**Enterprise / Adoption:**

- Fits real-world security expectations by combining privacy-by-design, policy checks, and controlled execution.
- Compatible with existing cloud LLM/VLM providers, making deployment easier without dependence on a single vendor.
- Creates a practical path for AI agents in privacy-sensitive domains where adoption is currently limited.

**Strategic / Long-term:**

- Can evolve from a browser extension into a reusable SDK or infrastructure layer for privacy-safe agents.
- Aligns directly with the SIH challenge by balancing context accuracy, PII detection, redaction quality, resource use, and latency.
- Positions privacy not as a restriction, but as an enabler for trustworthy autonomous assistance.

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# RESEARCH AND REFERENCES

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## ❖ References


1) **Browser & Runtime Foundations**

- SIH 26171 – On-device Visual Perception for Light-weight Browser Agents
- Chrome MV3 / Service Workers – https://developer.chrome.com/docs/extensions/develop/concepts/service-workers/lifecycle
- WebMCP – https://developer.chrome.com/docs/ai/agents
- MutationObserver – https://developer.mozilla.org/en-US/docs/Web/API/MutationObserver
- ONNX Runtime Web – https://onnxruntime.ai/docs/tutorials/web/

2) **Existing Systems Studied**

- Stagehand – DOM / Hybrid / CUA
  https://docs.stagehand.dev/v3/references/agent
- Agent-E – DOM distillation + change observation
  https://arxiv.org/abs/2407.13032
- Skyvern – secret placeholders + browser injection
  https://www.skyvern.com/docs/cloud/managing-credentials/credentials-overview
- OmniParser – screenshot → structured UI
  https://github.com/microsoft/OmniParser

3) **Privacy / OCR References**

- Microsoft Presidio – hybrid PII detection
  https://microsoft.github.io/presidio/analyzer/
- PaddleOCR Browser – local OCR
  https://www.paddleocr.ai/latest/en/version3.x/inference_deployment/cross_platform/browser.html
- UIDAI / Masked Aadhaar – https://uidai.gov.in/


## ◆ Research Workflow

Problem Statement → Requirement Analysis → Study Existing Systems → Feasibility & Threat Review → Final Architecture

## ❖ Comparison with Existing Systems

| Feature | OUR SYSTEM | Stagehand | Agent-E | Skyvern | OmniParser / UI-TARS |
|---|---|---|---|---|---|
| 1. Structured Browser Understanding | ✓ | ✓ | ✓ | ~ | ✗ |
| 2. Visual GUI Understanding | ✓ | ~ | ✗ | ✗ | ✓ |
| 3. Local PII Sanitization | ✓ | ✗ | ✗ | ~ | ✗ |
| 4. Incremental State / Change Tracking | ✓ | ~ | ✓ | ✗ | ✗ |
| 5. Blind Secret / PII Aliasing | ✓ | ✗ | ✗ | ✓ | ✗ |
| 6. Local Action Firewall | ✓ | ✗ | ✗ | ~ | ✗ |
| 7. Cloud-agnostic Reasoning | ✓ | ✓ | ~ | ~ | ~ |
| 8. Privacy-sensitive Workflow Fit | ✓ | ~ | ✗ | ~ | ✗ |
| 9. Purpose-bound Safe Context | ✓ | ✗ | ✗ | ✗ | ✗ |
| 10. Hybrid Structured + Vision Pipeline | ✓ | ~ | ✗ | ✗ | ✓ |

### Research Conclusion:

- Existing systems validate the building blocks: browser agents, vision parsing, local OCR, PII detection, and secret placeholders.
- Our solution is novel in combining structured-first perception, purpose-bound safe context, generalized blind aliasing, and a local action firewall.
- This makes the architecture more suitable for privacy-sensitive browser workflows than any single existing baseline.